CheckIn

Individual Contribution Report — Database & Backend

Student Attendance System · IDE201 Capstone · Group 2 · College of Science and Technology

Contributor	Ugyen Norbu (GitHub: UgayNobu)
Role	Database and backend; repository owner
Period covered	9 September – 11 October 2026
Repository	github.com/UgayNobu/Checkin (branch: dev)
Prepared for	Sprint report (compiled by Dechen)

1. Summary

This report lists the work I did myself on CheckIn. I set up the repository, designed and maintained the shared Supabase PostgreSQL database, connected it to the app through Prisma, and added the backend features from our design document in plain JavaScript. I also loaded public college data (departments, programmes, curriculum) and the AS2026 timetable, and built VLE-style module enrolment.

Work done on behalf of Sanskar (his frontend code and his attendance pages) is listed separately in Section 4 and is not counted as my contribution.

12	13	29	5	785
pull requests merged	commits	API routes added	database tables added	timetable slots loaded

2. Work completed

2.1 Repository and project setup

- **Repository structure:** created the repo with frontend, backend and docs folders, and the first README (PR #1, #2).
- **Project documents:** added the Group 2 proposal and the Sprint 1 report to docs/ and updated the Sprint 1 status and metrics (task T208).
- **Team workflow:** set up feature branches with pull requests into dev, so nobody pushes to dev directly. Kept .env files and secrets out of the public repo (.gitignore, PR #4).
- **Team guides:** wrote a setup guide (how to run the project, dos and don’ts) and an integration progress report for the team.

2.2 Database design (Supabase PostgreSQL)

- **Database:** designed the CheckIn database on Supabase, with readable IDs (U001, M001, P001 …) generated by gen_id() and sequences, check constraints, and row-level security enabled on every table.
- **Prisma:** created the Prisma schema from the live database and kept it in sync with every change (PR #3).
- **Test logins:** wrote scripts/set-test-passwords.js to set bcrypt password hashes for test users (PR #5).
- **Connecting the app:** replaced the mock database in the app with our real Supabase database. I adapted the schema mapping and wrote the SQL fixes so the existing app code worked against it (PR #6, database side).
- **One SQL file:** merged all SQL into backend/prisma/database.sql (structure plus seed data). It is safe to re-run and is applied in the Supabase SQL Editor. We never use prisma migrate or db push on the shared database (PR #7).
- **Roles and access:** set up separate database roles for student, tutor, admin and management. Created the admin (CST-ICT01) and management (CST-MGT01) accounts directly in SQL with bcrypt-hashed passwords, so these accounts cannot be self-registered.

2.3 Backend features from the design document (PR #7)

Our backend design document was written in TypeScript. As agreed with the team, I added its missing features to the existing server.js in plain JavaScript, without removing anything that already worked.

Area	What was added
Class sessions	List my sessions, end a session early, live roster with present / flagged / absent, distance and GPS accuracy
Flagged check-ins	Tutor accepts or rejects a flagged check-in; the student is notified
Check-in (GPS)	Saves position, distance and accuracy; flags outside-radius, low-accuracy and permission-denied check-ins
Exports	Module attendance as CSV and PDF (pdfkit); at-risk list and programme summary as CSV
Admin	Pending tutor approval (approve / reject), audit log viewer with filters, at-risk report
Alerts	Low-attendance job that notifies students below the required %, at most once every 7 days
Notifications	Unread count and mark all as read
Profile	Upload and remove a profile picture (PNG / JPG / WebP, max 2 MB)
Security	helmet security headers, a rate limit of 300 requests per minute, and a /health check

*Every route is documented in **backend/API_ADDITIONS.md**.*

2.4 Evidence documents and removing credits (PR #8)

- **Rename:** renamed relief_requests to evidence_documents throughout the database, Prisma, API and frontend labels. Existing data was kept.

- **Credits removed:** removed the credit system completely (tables, columns and code), as the team decided.
- **README:** rewrote the README to match the new design.

2.5 Departments and programmes (PR #9)

- **New table:** added a departments table with the 6 CST departments and their websites, all taken from cst.edu.bt.
- **Programmes:** added all 13 CST programmes and linked each to its department, with level and duration.
- **API and sign-up:** added /api/departments and /api/programs. The sign-up form now loads its dropdowns from these routes.

2.6 VLE-style module enrolment (PR #10)

- **Like Moodle:** a student picks a module, then enters that module's enrolment key.
- **Unique keys:** keys must be unique across modules (not case-sensitive).
- **Self-enrol switch:** the tutor can open or close self-enrolment for each module.
- **Remove a student:** a tutor or admin can unenrol a student, who is then notified.
- **Frontend:** updated the student Enrol screen and the tutor Modules page (self-enrol checkbox and Remove button).

2.7 Public curriculum and test data (PR #11)

- **Curriculum:** loaded the public curriculum for B.E. Software Engineering (27 modules) and B.E. Information Technology (35 modules) from the Computing Technologies Department website.
- **Shared modules:** added a program_modules table, so one module can belong to several programmes, years and semesters.
- **No real personal data:** test users are synthetic ("Test Student 01", @example.com addresses), because real student and staff data will only be given after the college approves the project.
- **Data sources:** added docs/DATA_SOURCES.md with APA references for every public source used.

2.8 Rooms and the AS2026 timetable (PR #12)

- **New tables:** added rooms and timetable_slots, and linked class_sessions to a timetable slot.
- **Data import:** parsed the college AS2026 timetable spreadsheet and loaded 37 rooms and 785 weekly class slots (theory, practical and tutorial) with rooms and tutors. We used this data with our guide's permission.
- **API:** added /api/rooms, /api/timetable, /api/student/timetable and /api/teacher/timetable.
- **Privacy:** the timetable data file contains staff names, so it is kept out of the public repository and shared only within the team.

Database item	Count	Source
Departments	6	cst.edu.bt
Programmes	13	cst.edu.bt
Curriculum modules (SWE + IT)	62	ctd.cst.edu.bt
Modules in total	168	Curriculum + AS2026 timetable

Database item	Count	Source
Rooms	37	AS2026 timetable
Timetable slots	785	AS2026 timetable
Test users	8 + 2 staff accounts	Synthetic

3. Good practices followed

- Secrets (database password, JWT secret) are never committed. .env.example holds only placeholders.
- All database changes go through one re-runnable SQL file, tested on a local PostgreSQL copy before running it on Supabase.
- Every change went through a feature branch and a pull request into dev.
- New features were added alongside existing code, never replacing teammates' working features.
- Public data is referenced. Synthetic data is used for people until the college approves real data.

4. Not counted as my contribution

I helped with the following, but the work belongs to Sanskar, so it is not counted above:

- Sanskar's frontend and Express backend code, which I pushed to the repository for him in PR #6.
- His attendance codes, flagged check-ins, teacher records and reports, and admin programme pages (PR #13). I only resolved the merge with dev.
- Account fixes for Sanskar (password reset, removing a duplicate account).

5. Next steps

- Timetable screens for students and tutors, with "Start session" from a timetable slot.
- Link tutor accounts directly to their timetable slots.
- Add room coordinates and widen the geofence from 10 m to 25–50 m for indoor GPS.
- Deploy the app so it works on phones and on the college Wi-Fi. The college network blocks the database ports.
- Before deployment, change the database password and JWT secret.

Appendix: My commits

Date	PR	Commit	Description
2026-09-09	#1	7529267	Add README and project folder structure (T208)
2026-09-09	#1	679880b	Add proposal and Sprint 1 report to docs (T208)
2026-09-09	#1	7cc02ae	Update Sprint 1 report status and metrics (T208)
2026-10-06	#2	09b0161	Update README
2026-10-07	#3	9993bcc	Add Prisma schema from Supabase
2026-10-08	#4	f7f1da6	Ignore .DS_Store

Date	PR	Commit	Description
2026-10-08	#5	9dabed1	Script to set bcrypt hashes for test users
2026-10-09	#6	e83f7f5	Connect app to Supabase database (database side only)
2026-10-09	#7	7bc7274	Backend features from design doc; single database.sql
2026-10-10	#8	29d6906	Evidence documents rename, remove credits, new README
2026-10-10	#9	0858b36	Departments table and all CST programmes
2026-10-10	#10	0aa2321	VLE-style enrolment: keys, self-enrol switch, unenrol
2026-10-10	#11	957f44c	Public SWE/IT curriculum, program_modules, test data, sources
2026-10-10	#12	2dd4328	Rooms and AS2026 timetable with timetable API